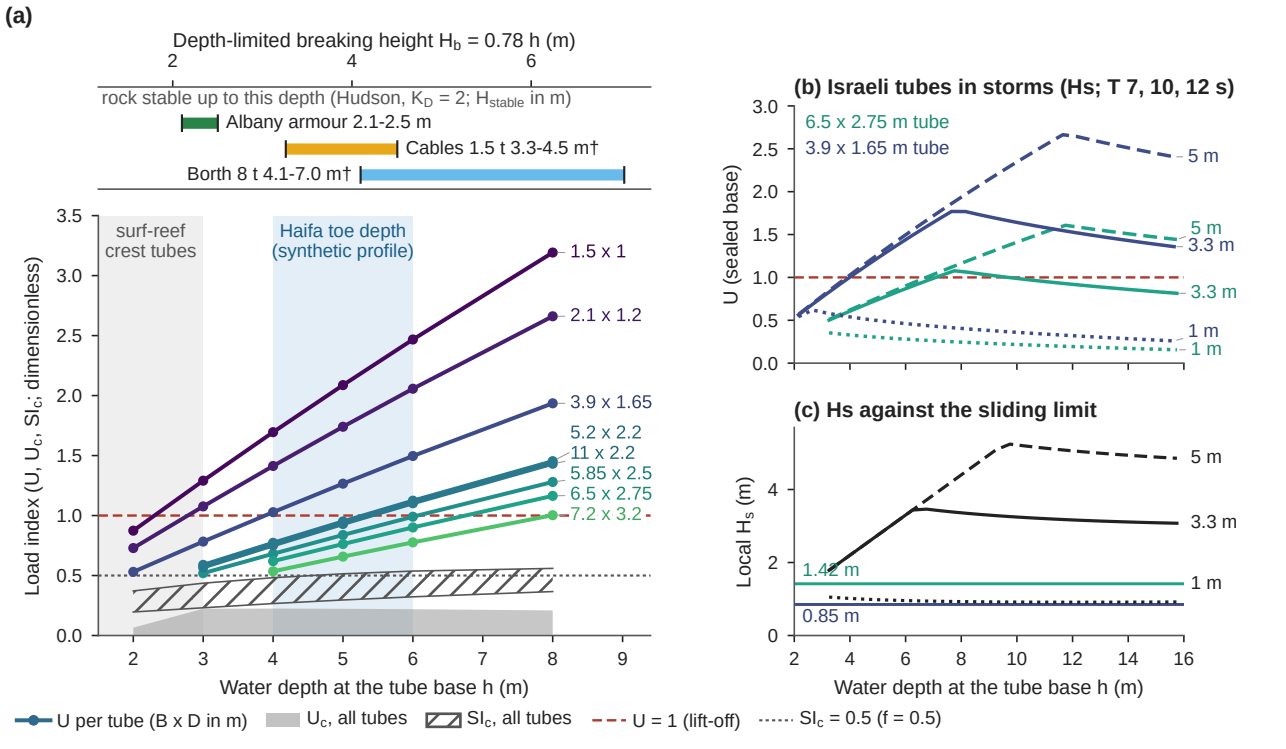
(a)
Depth-limited breaking height H_b = 0.78 h (m)
2
4
6
rock stable up to this depth (Hudson, K_D = 2; H_{stable} in m)
Albany armour 2.1-2.5 m
Cables 1.5 t 3.3-4.5 m†
Borth 8 t 4.1-7.0 m†
surf-reef crest tubes
Haifa toe depth (synthetic profile)
1.5 x 1
2.1 x 1.2
3.9 x 1.65
5.2 x 2.2
11 x 2.2
5.85 x 2.5
6.5 x 2.75
7.2 x 3.2
Load index (U, U_c, SI_c; dimensionless)
Water depth at the tube base h (m)
(b) Israeli tubes in storms (Hs; T 7, 10, 12 s)
6.5 x 2.75 m tube
3.9 x 1.65 m tube
5 m
5 m
3.3 m
3.3 m
1 m
1 m
U (sealed base)
(c) Hs against the sliding limit
5 m
3.3 m
1 m
1.42 m
0.85 m
Local H_s (m)
Water depth at the tube base h (m)
U per tube (B x D in m)
U_c, all tubes
SI_c, all tubes
U = 1 (lift-off)
SI_c = 0.5 (f = 0.5)